Base station physics: summary

Params: base/spec/params.py v1.5.0

Spring (each, A228, SF 1.2 on Shigley static limits)

Rate used / by formula	0.125 / 0.117 N/mm
Initial tension used	7.0 N
Body coils, mass	98, 53 g
Safe force (governed by)	40.9 N (hook torsion)
Safe extension (wire) / spec cap	271 / 265 mm
Surge (fixed-fixed)	24 Hz

Launch (chain spring model)

Stroke, ratio, pitch	241 mm, 1.0:1, 35 deg
Spring extension full / rest	265 / 24 mm
Stored energy (2 springs)	12.1 J
Carriage force at full draw	80 N
Muzzle speed	6.87 m/s
Peak acceleration, stroke time	18 g, 59 ms
Efficiency (dart KE / stored)	55 %
Carriage KE at stop, pad peak	1.90 J, 287 N (19 mm pad)
Spring KE at stop, coil clash	0.83 J, 8.3 m/s

Flight (bead-chain tether + reel model, level ground)

Line path at the shot	taut (path only lengthens)
Range at 0.25 / 0.5 / 0.75 N brake	4.79 / 4.25 / 3.77 m (pitch 38)
Same shot, no tether	5.16 m
Flight time, apex (0.5 N)	0.91 s, 1.32 m
Impact speed, angle (0.5 N)	6.2 m/s, 55 deg down
Ballistics table	6 to 47 deg, best 38.5 deg: 4.25 m

Storage reel payout (LIGHT 0.25 N at full reel)

Reel empty mass, inertia full	98 g, 26.2 kg mm2
Line radius full / core	27.4 / 20.0 mm (brake +37 % at core)
Line pull in the stroke (spin-up)	4.82 N
Reel rpm max	2388 rpm
Overrun after landing	1.37 m in 0.59 s

Capstan and tow

Wraps for tow limit at the capstan (59 N, mu 0.15)	2.6 raw, 4 design
Take-up needed for the tow at the actual wraps (7)	0.08 N
Tow at limit: torque, speed, current	3.27 N m, 0.142 m/s, 5.10 A
Wiper at limiter (5 N m), 12.6 V: torque, current, line pull	5.0 N m, 7.0 A, 79 N (break 667 N)
Wiper motor stall (no limiter), 12.6 V: torque, current, line pull	14.1 N m, 17.0 A, 223 N
Tow sensing at the dart: N per wiper amp above free	dart = 14.4 x (I - I_free) - 1.8 N (capstan side x1.18)

Drivetrain (wiper at 11.1 V with sag)

aim: peak torque, current, time	0.33 N m, 1.9 A, 7 s
cock: peak torque, current, time	2.93 N m, 4.7 A, 3 s
reel: peak torque, current, time	0.38 N m, 1.9 A, 106 s
tow: peak torque, current, time	3.27 N m, 5.1 A, 35 s

To measure (then edit base/spec/params.py and run ./run_all.sh)

1. Spring hang test, each spring: hang 1, 2, 3 kg and measure the length each time (hooks included). python calib.py spring FREE_MM 1:MM 2:MM 3:MM, then SPRING k_n_per_mm and F0_n (mean of the two); the stronger spring decides the anchor holes (MEASURE_LIST B4). Every range number scales with the stored energy.
2. Wiper (MEASURE_LIST A6): free speed and current from a 40 % bench jog in telemetry, then WIPER n0_rpm = jog rpm x 2.5 (never the raw jog rpm) and free_a = amps. The stall cannot be read (jogs trip at 2.5 to 3 A, A7): keep stall_nm and stall_a.
3. Dart mass on a kitchen scale with ballast and tether knot (spec 0.279 kg), then DART mass_kg.
4. Payout brake (MEASURE_LIST G4, hanging weights: a luggage scale cannot read this low): in LAUNCH 20 g on the line must not pull line off and 30 g must (LIGHT 0.25 N); in NEUTRAL the FIRM clutch (5 N) lets line off at about 0.5 kg on the luggage scale.
5. First shots on level ground: python calib.py range pitch:brake:range ..., then LAUNCH v_scale = FACTOR.